Lecture 10: Bond Basics and Valuation

Questions

These five multiple-choice questions follow the bond-pricing workflow used in Lecture 10, with different numbers. All bonds pay semiannually. Assume no default, fees, taxes or early redemption, and value the bonds on a coupon date. Annual spot rates and YTM are quoted with semiannual compounding. Keep full precision until the final answer.

Bond A, Questions 1-4: face value $2,000; annual coupon rate 8%; maturity 2 years; two coupon payments per year. Its annual spot curve is shown below.

Maturity (years)	0.5	1.0	1.5	2.0
Annual spot rate	9.00%	9.50%	10.00%	10.50%

1. Identify every cash flow

Which sequence lists all four future cash flows at 0.5, 1.0, 1.5 and 2.0 years?

A. $160, $160, $160, $2,160

B. $80, $80, $80, $2,080

C. $80, $80, $80, $2,000

D. $40, $40, $40, $2,040

2. Price using the spot curve

Discount every cash flow at its matching spot rate. What is Bond A worth today?

A. $1,913.59

B. $2,000.00

C. $1,929.08

D. $2,057.25

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3. Compute quoted annual YTM in Excel

Use Bond A's price of $1913.59151439. Which Excel formula and quoted annual YTM are correct? Format the Excel result as a percentage.

A. =RATE(4,80,-1913.59151439,2000); 5.2248%

B. =2*RATE(4,80,-1913.59151439,2000); 10.4496%

C. =(9%+9.5%+10%+10.5%)/4; 9.7500%

D. =2*RATE(2,160,-1913.59151439,2000); use two annual payments

4. Above, at or below par?

Consider the same coupon contract at three alternative market YTMs. Scenario I: 6%; Scenario II: 8%; Scenario III: 10%. Which ordering correctly classifies the bond?

A. Discount; par; premium

B. Premium; discount; par

C. Premium; par; discount

D. Par; premium; discount

5. Repeat the full workflow with a lower spot curve

Bond B has the same $2,000 face value, 8% coupon and two-year maturity, but annual spot rates are 5%, 5.5%, 6% and 6.5% at 0.5, 1, 1.5 and 2 years. Calculate the price using all four cash flows, then compute quoted annual YTM with Excel. Which result is correct?

A. $2,057.25; 3.2257%; premium

B. $1,913.59; 10.4496%; discount

C. $2,057.25; 6.4514%; discount

D. $2,057.25; 6.4514%; premium

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Worked solutions

1. Every cash flow - Answer B

Step 1. I calculate the coupon from face value, not the purchase price. The annual coupon is $160; two payments per year means $80 each. Two years gives four payment dates.

$$C = \frac{2{,}000 \times 0.08}{2} = \$80, \qquad N = 2 \times 2 = 4$$

Step 2. Principal is returned only at maturity. The last payment includes principal and the final coupon. All four future payments are therefore:

$$CF_{0.5} = 80, \quad CF_{1.0} = 80, \quad CF_{1.5} = 80, \quad CF_{2.0} = 2{,}080$$

2. Spot-discounted price - Answer A

Step 1. Divide each annual spot rate by two. Use one half-year period for the first payment, two for the second, three for the third and four for the last. Each spot rate discounts its entire matching maturity; we do not multiply successive spot rates.

$$P_0 = \frac{80}{1.045} + \frac{80}{1.0475^2} + \frac{80}{1.05^3} + \frac{2{,}080}{1.0525^4}$$

Maturity (years)	Spot rate	Cash flow ($)	Discount factor	PV ($)
0.5	9.00%	80.00	0.956938	76.5550
1.0	9.50%	80.00	0.911364	72.9091
1.5	10.00%	80.00	0.863838	69.1070
2.0	10.50%	2,080.00	0.814914	1695.0204

Step 2. Add all four present values, including the principal inside the last cash flow. The discount factor is just a multiplier; the price comes from adding the discounted cash flows.

$$P_0 = \sum_{k=1}^{4} CF_k (1 + s_k/2)^{-k} = \boxed{\$1,913.5915}$$

In Excel, with maturity in A21:A24, annual spot rates in B21:B24 and cash flows in C21:C24, use **=1/(1+B21/2)^(2*A21)** in D21 and **=C21*D21** in E21. Copy through row 24 and add **=SUM(E21:E24)**. Enter rates as percentages, such as 9%.

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3. YTM in Excel - Answer B

Step 1. We now know the price. YTM is the single rate that reproduces that price using the same four cash flows. Let i be the six-month yield. Nothing changes in the coupon or principal payments.

$$1913.591514 = \frac{80}{1+i} + \frac{80}{(1+i)^2} + \frac{80}{(1+i)^3} + \frac{2{,}080}{(1+i)^4}$$

Step 2. Enter **=RATE(4,80,-1913.59151439,2000)**. The inputs are four periods, $80 per period, a negative purchase price and $2,000 principal. Do not put $2,080 in RATE's final-value argument: RATE already includes the final $80 coupon.

$$i = 5.224779\%, \qquad YTM = 2i = \boxed{10.449558\%}$$

Step 3. The complete Excel formula is **=2*RATE(4,80,-1913.59151439,2000)**. Format as Percentage. RATE gives the six-month rate; multiplying by two gives the annual quote requested in class. Averaging the spot rates does not calculate YTM.

You can also enter the complete investment cash-flow series in five consecutive cells: negative price at time zero, then 80, 80, 80 and 2,080. **=2*IRR(the_five_cells)** gives the same annual quoted YTM because those observations are equally spaced six months apart.

4. Premium, par and discount - Answer C

Step 1. The coupon rate remains 8% in all three scenarios. Compare that fixed coupon rate with the market YTM. A lower required yield makes the same promised payments worth more; a higher required yield makes them worth less.

Scenario / annual YTM	Price ($)	Compared with $2,000
I: 6%	2,074.34	Above par: premium
II: 8%	2,000.00	At par
III: 10%	1,929.08	Below par: discount

Step 2. These prices use the same $80, $80, $80 and $2,080 payments, discounted at one-half of each annual YTM. You can verify them in Excel with **=-PV(6%/2,4,80,2000)**, then replace 6% with 8% and 10%. Bond A itself trades below par because its spot-discounted price is below $2,000 and its YTM is above 8%.

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5. Full workflow for Bond B - Answer D

Step 1. Keep the contractual cash flows. Only the market spot curve changes. The coupon is still \$80 every six months, there are still four payments, and the final payment is still \$2,080. Lower market rates do not reduce the promised coupon.

$$P_B = \frac{80}{1.025} + \frac{80}{1.0275^2} + \frac{80}{1.03^3} + \frac{2{,}080}{1.0325^4}$$

Maturity (years)	Spot rate	Cash flow (\$)	Discount factor	PV (\$)
0.5	5.00%	80.00	0.975610	78.0488
1.0	5.50%	80.00	0.947188	75.7751
1.5	6.00%	80.00	0.915142	73.2113
2.0	6.50%	2,080.00	0.879913	1830.2191

$$P_B = \boxed{\$2,057.254323}$$

Step 2. Solve YTM from the price. Use the price you just calculated, not face value. Enter the Excel formula below, keeping the price negative because it is your cash outflow today.

=2*RATE(4,80,-2057.25432332,2000)

$$i_B = 3.225691\%, \qquad YTM_B = 2i_B = \boxed{6.451382\%}$$

Step 3. Check using every cash flow. Insert this single YTM into the pricing equation. Its present value should match the price obtained from the spot curve. The individual spot-discounted values need not match those from YTM; their totals match.

$$P_B = \frac{80}{1 + YTM_B/2} + \frac{80}{(1 + YTM_B/2)^2} + \frac{80}{(1 + YTM_B/2)^3} + \frac{2{,}080}{(1 + YTM_B/2)^4}$$

Step 4. Interpret. Bond B trades above \$2,000 face, so it is a premium bond. Its 8% coupon rate is above its YTM, which confirms the classification. Compare it with Bond A: identical cash flows, different discount rates, different prices. That is the main valuation idea in this lecture.